

Project: FetReg

Placental Vessel Segmentation and Registration in Fetoscopy

Group members

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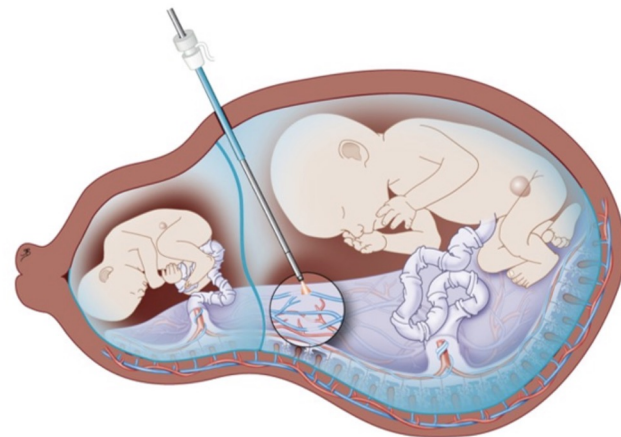
Supervisors

Sophia Bano & Francisco Vasconcelos

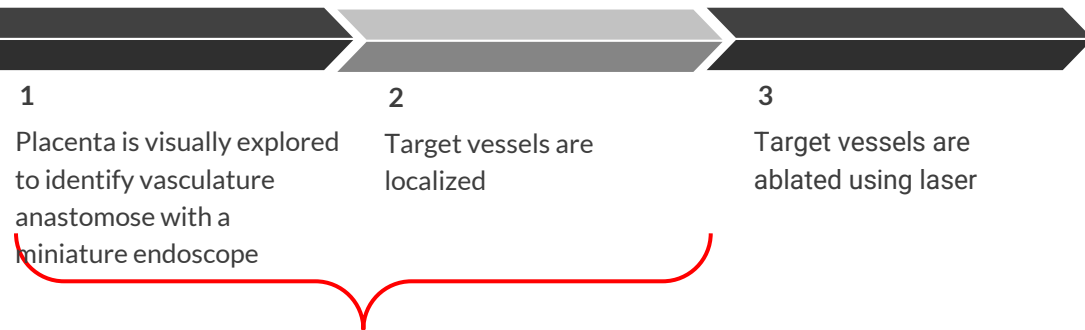
University College London



- Fetal anomaly affecting identical twins sharing a **monochorionic placenta** (10-15% of monochorionic twins)
- Blood transferred from one twin to the other via vascular anastomosis in the placenta
- **Dangerous for both twins**
 - Donor experience much slower growth
 - Recipient at risk of heart failure
- **Treatment**
 - Laser of vascular anastomosis (**photocoagulation**)
 - Two separate placentas

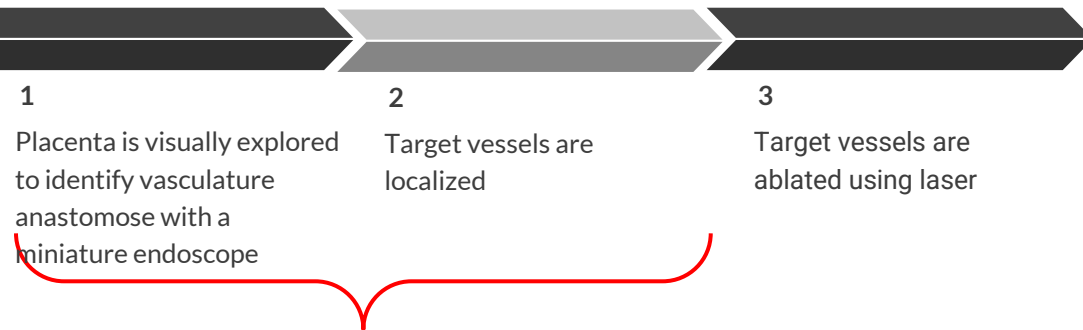


Fetoscopic Laser Photocoagulation



- Limited field-of-view
- Difficult visual conditions
 - *Poor visibility*
 - *Specular highlights*
 - *Occlusions*
- Dynamically changing views
- Ground-truth not available

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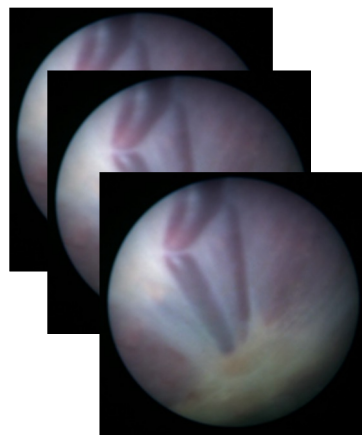


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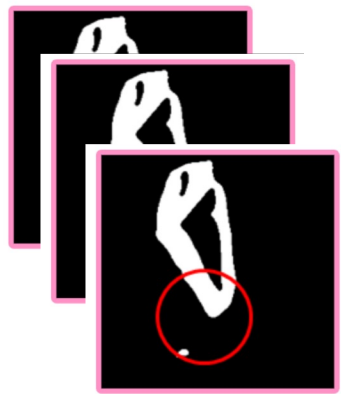


Automatic **segmentation** of placental vessels
and identification of
anastomotic ones
&
Registration of vessels images
for an expanded vessel map

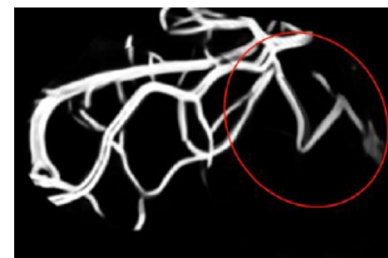
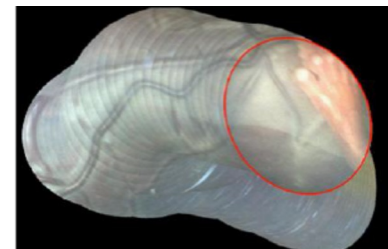




Segmentation
model



Registration
model



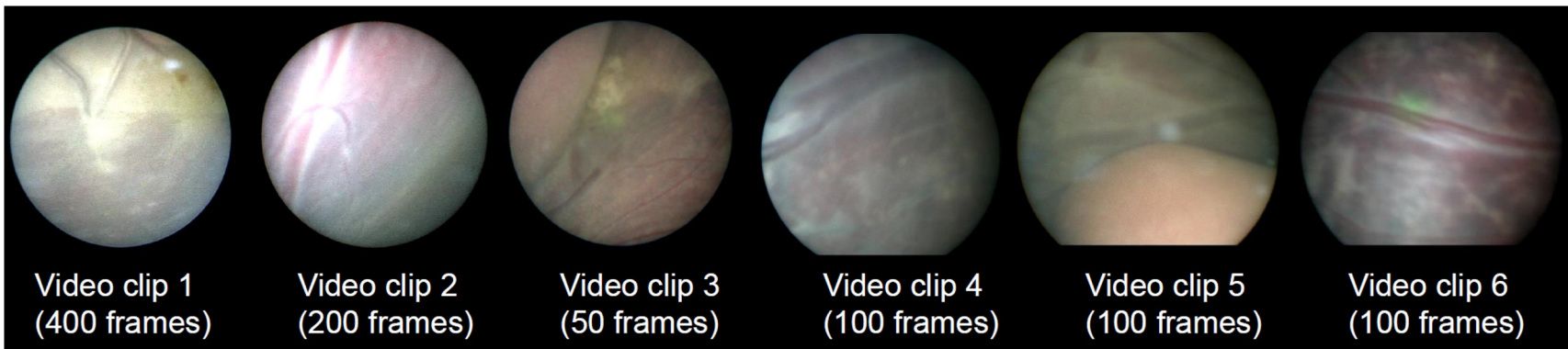
Description

1. Implement Segmentation Architecture (other than Unet) for placental vessel segmentation
2. Train the architecture for Fold 1 only (because of time limitation)
3. Compare the results with the baseline ones for Fold 1

Code

[Colab Notebook](#)

- The dataset includes 6 fetoscopic video clips (950 frames)
- **Train / Valid/ Test:** Train **02, 03, 04, 05**; Valid: **06**; Test: **01**
- Preprocessing:
 - 1) Convert 3 channel mask into single channel gray image by using
$$mask = cv2.imread(self.masks_fps[i], 0)$$
 - 1) The dimensions of the frames are not the same. Thus, we need resize the frame and corresponding mask.

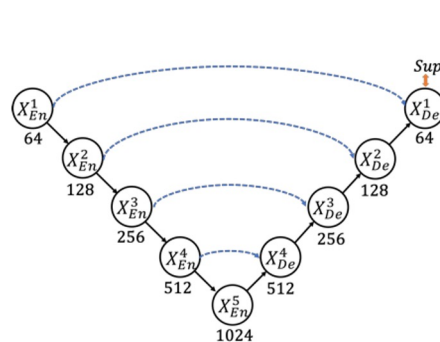


● Model

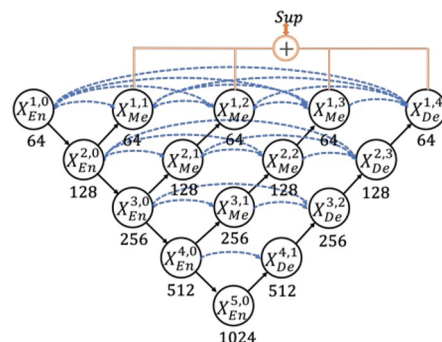
- Unet
- Unet++
- DeepLabV3

● Backbone

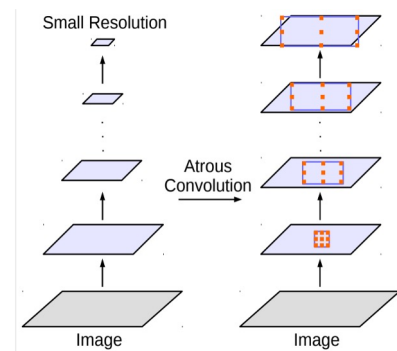
- Vgg16
- Resnet50
- Resnet101



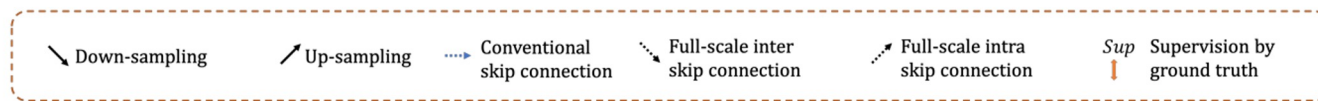
(a) UNet
(Plain skip connections)



(b) UNet++
(Nested and dense skip connections)



(c) DeepLabV3
(Dilated Convolution)



● Loss function

$$Dice\ loss = 1 - \frac{2|X \cap Y|}{|X| + |Y|}$$

● Evaluation Metrics

$$Dice = \frac{2|X \cap Y|}{|X| + |Y|} = \frac{2 \cdot TP}{(TP + FN) + (TP + FP)}$$

$$IoU = \frac{|X \cap Y|}{|X \cup Y|} = \frac{TP}{TP + FN + FP} = \frac{Dice}{2 - Dice}$$

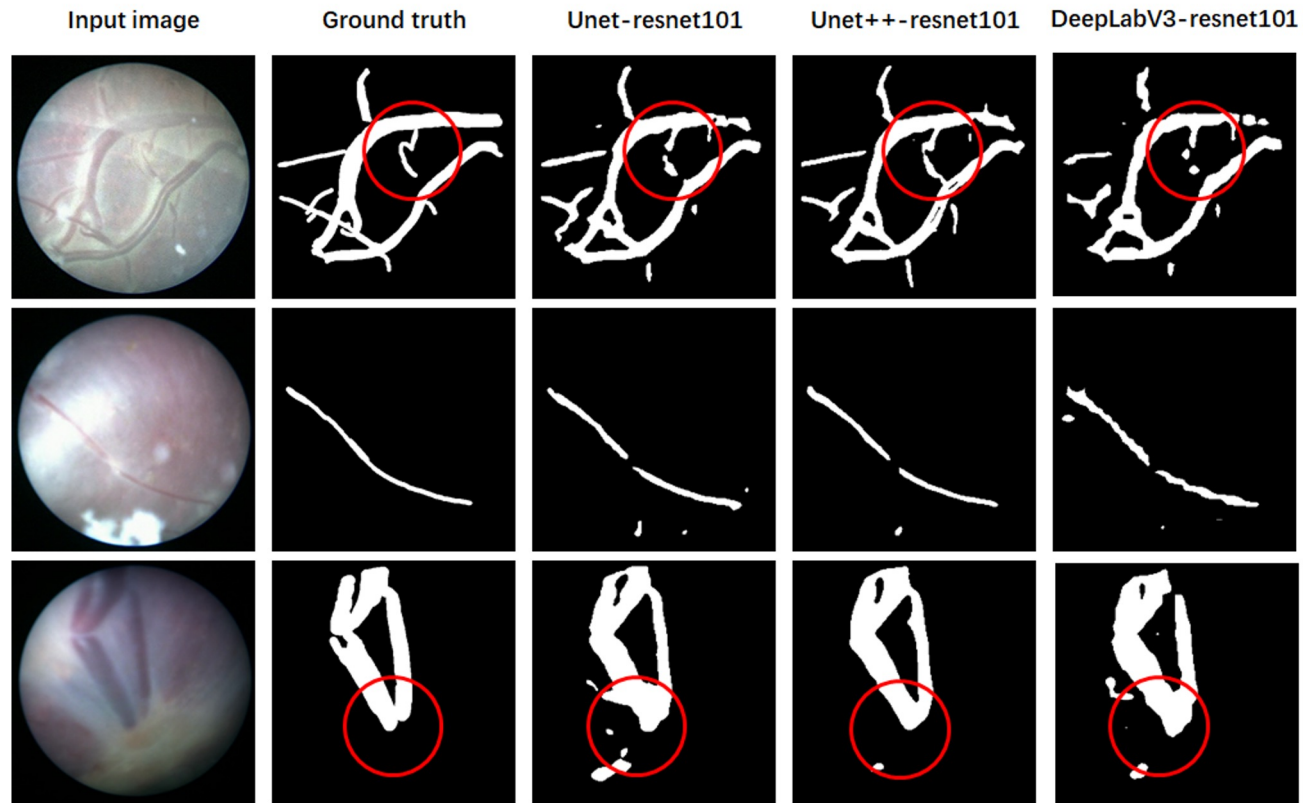
$$Sensitivity = \frac{TP}{TP + FN}$$

$$PPV = \frac{TP}{TP + FP}$$

Quantitative Comparison

Model	Encoder	Trainable params	Dice	IoU	Sensitivity	PPV
Unet	vgg16	23,748,241	81.49 ± 9.03	69.54 ± 10.47	81.85 ± 10.88	83.26 ± 10.19
	Resnet50	32,521,105	80.66 ± 7.74	68.25 ± 10.23	83.69 ± 10.58	79.54 ± 10.86
	Resnet101	51,513,233	82.55 ± 6.78	70.81 ± 9.20	84.89 ± 9.43	81.50 ± 9.13
Unet++	vgg16	34,626,705	82.26 ± 10.20	70.83 ± 11.36	85.53 ± 12.47	81.41 ± 9.70
	Resnet50	48,985,745	82.72 ± 7.58	71.19 ± 10.14	84.21 ± 10.53	82.52 ± 9.06
	Resnet101	67,977,873	84.33 ± 6.06	73.36 ± 8.68	84.10 ± 9.09	85.50 ± 7.72
DeepLabV3	Resnet50	39,633,729	78.49 ± 8.49	65.35 ± 10.82	80.99 ± 10.52	77.35 ± 10.47
	Resnet101	58,625,857	78.49 ± 8.34	65.33 ± 10.75	80.31 ± 11.28	78.21 ± 10.31

Visual Comparison



- Implement **Unet**, **Unet ++** and **DeepLabV3** with **different backbones**
- **Unet ++** (Resnet 101) generally outperforms other structures compared
- Train the model with **Dice loss rather than binary cross-entropy** [1]
- Evaluate the model with **additional metrics**, e.g. Dice, IoU, sensitivity, PPV

Future work

- Try other novel architectures (**TransUNet** [3])
- Train the model with more epochs
- Add data augmentation to increase the robustness
- Add learning rate decay to help both optimization and generalization

Description

1. Implement a direct pairwise consecutive frames registration method using either RGB or vessel maps (or both) provided in the sequence data
2. Use the provided Registration visualization code to generate the mosaic for at least video 1 sequence

Code

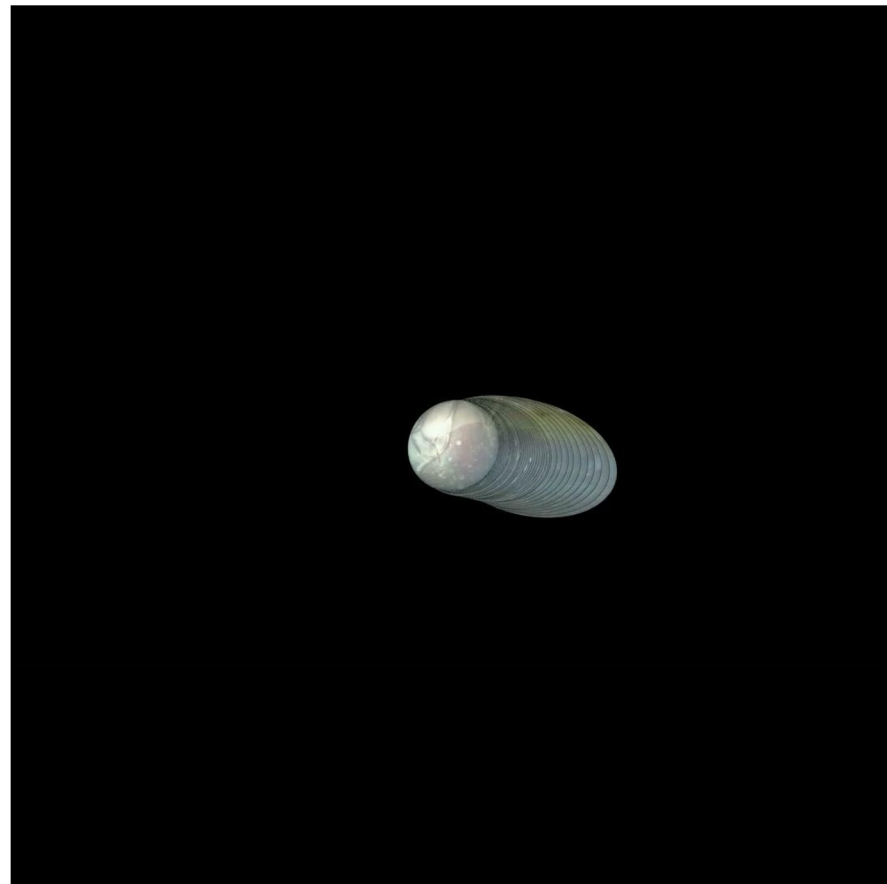
[GitHub Repository](#)

- It is assumed that the mapping from one image to the next is an **affine transformation**
- Images show many sorts of problems. Low contrast, noise (ought to amniotic fluid), etc. We use the **segmentation masks** for the estimation.
- We tried **two approaches**:
 - Estimating the mapping directly from segmentation masks ([AIRLAB](#))
 - Estimating the mapping from the deformation field estimated with optical flow on gray values

Sample Sequence



Sample Sequence



- Estimating homography from optical flow is not reliable
- Using a data-driven approach (deep learning) that learns robust features from data should yield better results
- Multi-task learning: Segmentation + Registration

Using an approach that simultaneously learns to segment and register frames could be the most promising approach

- [1] Bano, S., Vasconcelos, F., Shepherd, L.M., Vander Poorten, E., Vercauteren, T., Ourselin, S., David, A.L., Deprest, J. and Stoyanov, D., 2020, October. Deep placental vessel segmentation for fetoscopic mosaicking. In International Conference on Medical Image Computing and Computer-Assisted Intervention (pp. 763-773). Springer, Cham.
- [2] Bano, S., Casella, A., Vasconcelos, F., Moccia, S., Attilakos, G., Wimalasundera, R., David, A.L., Paladini, D., Deprest, J., Mattos, L.S. and Stoyanov, D., 2021. FetReg: Placental Vessel Segmentation and Registration in Fetoscopy Challenge Dataset. arXiv preprint arXiv:2106.05923.
- [3] Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L. and Zhou, Y., 2021. Transunet: Transformers make strong encoders for medical image segmentation. arXiv preprint arXiv:2102.04306.
- [4] Code base: [GitHub - psinger/segmentation_models.pytorch: Segmentation models with pretrained backbones. PyTorch.](https://github.com/psinger/segmentation_models.pytorch)



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UCL Medical Image Computing Summer School
(MedICSS), 2021

